

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

Expected May 2028

College Park, MD

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Building capacity planning for Amazon Redshift maintenance like patching: Java logic that sets upper and lower bounds on reserved EC2 capacity to balance availability against cost, validated by backtesting

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Cut LLM workflow generation time **65%** to under 15s by splitting one large Claude skill into smaller sub-skills that call Python tools for search and validation, and output quality held in repeated stress-test runs
- Precomputed popular workflows from a 270K+ row PostgreSQL table into a category tree that the Claude skill searches first, with Redis caching full workflows, so lookups rarely touch a database gated by permissions and rate limits
- Rebuilt the legacy tool developers use to create test-data workflows as a React and TypeScript Electron app with a visual React Flow editor and Jest-tested validation, and **90%** of 30+ surveyed developers preferred it

UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Automated retraining for a production Amazon Bedrock classifier that rates research documents by importance: on an EventBridge schedule a larger LLM grades its ratings, and fine-tuning runs only when the grades shift past a threshold
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Co-built a research tool that scores and aggregates topics so Amazon Leo's public-policy staff can see what's popular, running in dev on AWS Fargate with GitHub Actions CI/CD and AWS CDK infrastructure-as-code ready for production

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- For a PhD-led robotics project aimed at ICLR, built a pipeline that overlays AnyGrasp's candidate grasps on camera images in Open3D for a vision-language model, then tuned prompts so it favors grasps that avoid hitting the table

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Building a RAG meal-suggestion feature for the official UMD app, in which GPT-4o mini writes suggestions from scraped recipes found by SentenceBERT vector search in ChromaDB, and internal testers rated **93%** of meals satisfactory
- Built Redis caching in front of the Prisma ORM queries for Tortuga, the studio's class scheduler with 20,000+ visits in its first 30 days, to reduce database load and latency

Projects

IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine and replayed **50.9M** real IEX exchange messages through it without ever resyncing to the exchange's book, agreeing with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices

Technical Skills

AI/ML: LLMs (Claude, Amazon Bedrock, GPT-4o mini), LLM agents and tool use (Claude skills), retrieval-augmented generation (RAG), embeddings and vector search (SentenceBERT, ChromaDB), LLM-as-a-judge evaluation, model fine-tuning, vision-language models, prompt engineering, Open3D

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL

Cloud & Data: AWS (Lambda, Kinesis, EventBridge, Fargate), AWS CDK, monitoring (CloudWatch, X-Ray), REST APIs, PostgreSQL, Redis, OpenSearch

Engineering: full-stack apps (React, Electron), data structures and algorithms, object-oriented design, Docker, GitHub Actions (CI/CD), Linux, unit testing (Pytest, Jest), Git, uv

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)